Response Times Models

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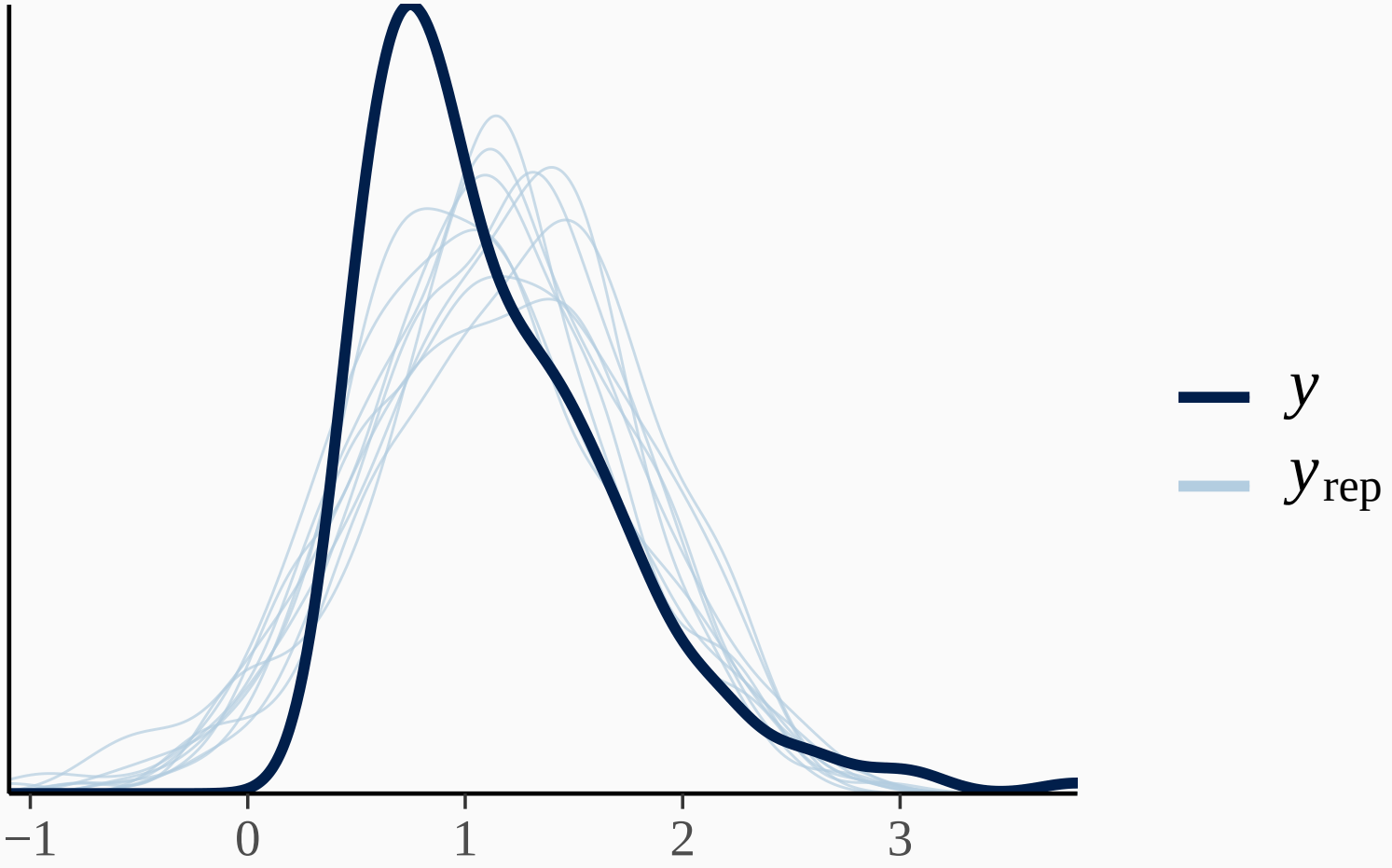
Motivation

- RTs are very common in cognitive sciences
- Rarely (if ever) normally distributed
- Often paired with dichotomous response data
- Nice statistical models available

Response Times and the Normal Distribution

```
fit_rt1 <- brm(rt ~ grp, data = RT)
```



The ExGaussian Distribution

The ExGaussian (short for exponentially modified Gaussian) distribution shows much better fit to respone time data than the normal distribution

If $X \sim \mathcal{N}(\xi, \sigma)$ and $Y \sim \mathrm{Exp}(\beta)$ then

$$X + Y = Z \sim \mathrm{ExGaussian}(\xi, \sigma, \beta)$$

The parameter β controls the right-skewness of the distribution.

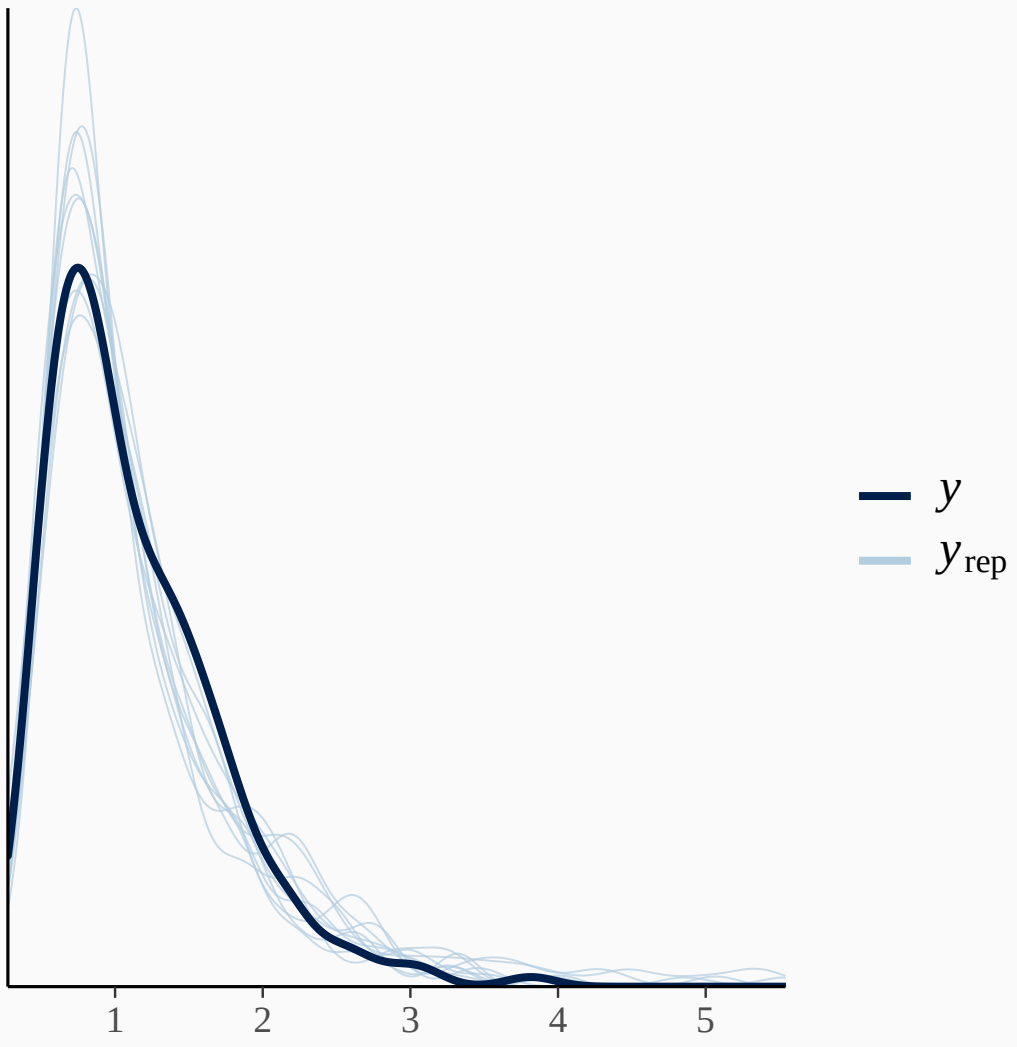
Since $\mu := \mathrm{E}(Z) = \xi + \beta$, we predict μ instead of ξ.

The ExGaussian Distribution: Example

```r
fit_rt2 <- brm(rt ~ grp, data = RT, exgaussian())
```

```
##  Family: exgaussian
##   Links: mu = identity; sigma = identity; beta = identity
## Formula: rt ~ grp
##    Data: RT (Number of observations: 200)
## Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup samples = 4000
##
## Population-Level Effects:
##           Estimate Est.Error l-95% CI u-95% CI Eff.Sample Rhat
## Intercept     1.15      0.05     1.06     1.26       1979 1.00
## grpB         -0.03      0.04    -0.11     0.04       2772 1.00
##
## Family Specific Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Eff.Sample Rhat
## sigma     0.09      0.03     0.04     0.17       2287 1.00
## beta      0.63      0.06     0.53     0.75       1749 1.00
##
## Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
## is a crude measure of effective sample size, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

The ExGaussian Distribution: Visualization

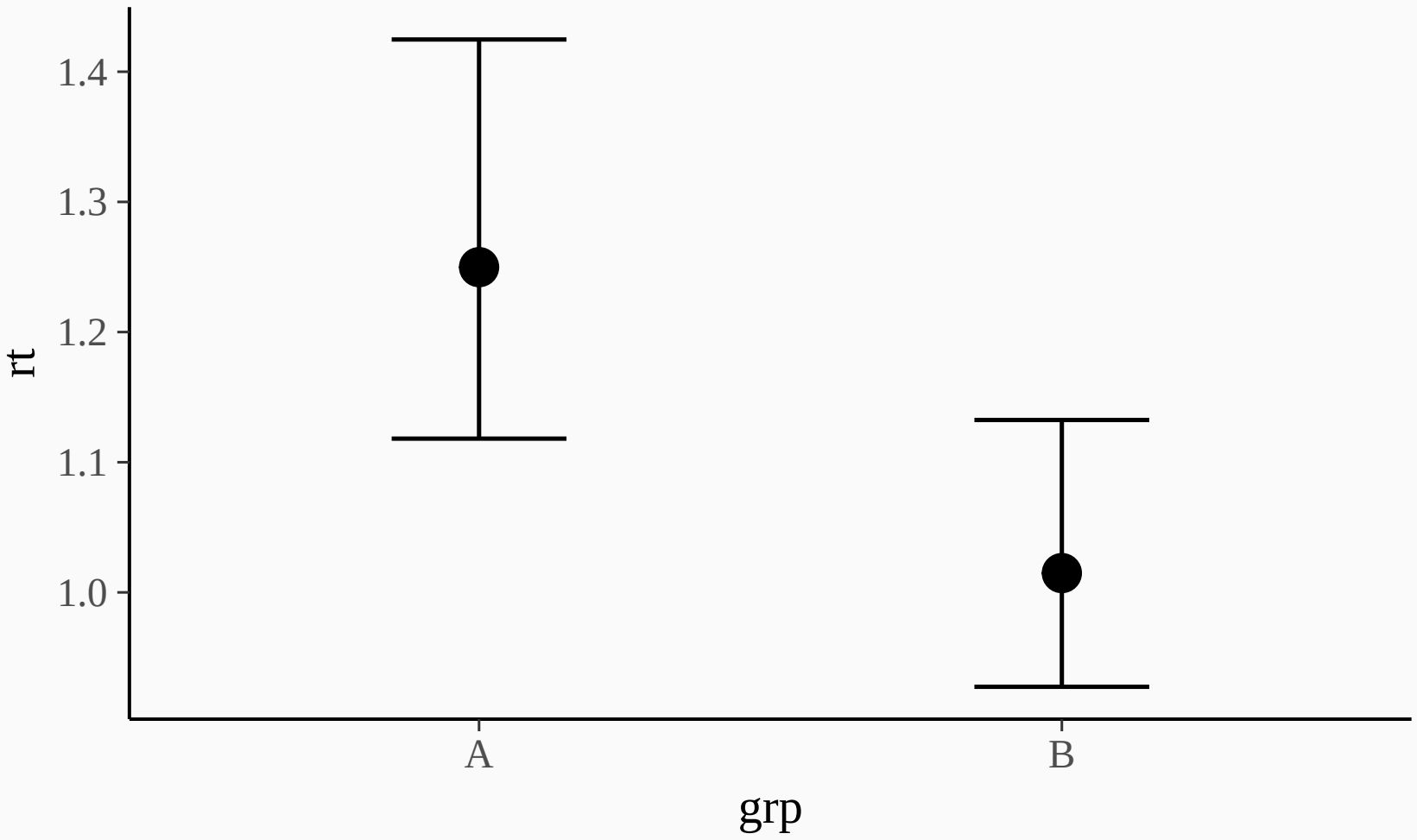


The ExGaussian Distribution: Predicting All Parameters

```r
prior <- c(prior(normal(0, 5), dpar = "sigma"),
           prior(normal(0, 5), dpar = "beta"))
fit_rt3 <- brm(bf(rt ~ grp, sigma + beta ~ 0 + grp), data = RT,
               family = exgaussian(), prior = prior)
```

```
##  Family: exgaussian
##   Links: mu = identity; sigma = log; beta = log
## Formula: rt ~ grp
##          sigma ~ 0 + grp
##          beta ~ 0 + grp
##    Data: RT (Number of observations: 200)
## Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup samples = 4000
##
## Population-Level Effects:
##            Estimate Est.Error l-95% CI u-95% CI Eff.Sample Rhat
## Intercept      1.26      0.08     1.12     1.42       1461 1.00
## grpB          -0.24      0.09    -0.43    -0.06       1656 1.00
## sigma_grpA    -2.95      0.64    -4.16    -1.56       1532 1.01
## sigma_grpB    -2.29      0.30    -2.91    -1.75       2705 1.00
## beta_grpA     -0.27      0.12    -0.52    -0.04       1344 1.00
## beta_grpB     -0.70      0.12    -0.93    -0.45       2175 1.00
##
## Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
## is a crude measure of effective sample size, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

The ExGaussian Distribution: Predicting All Parameters



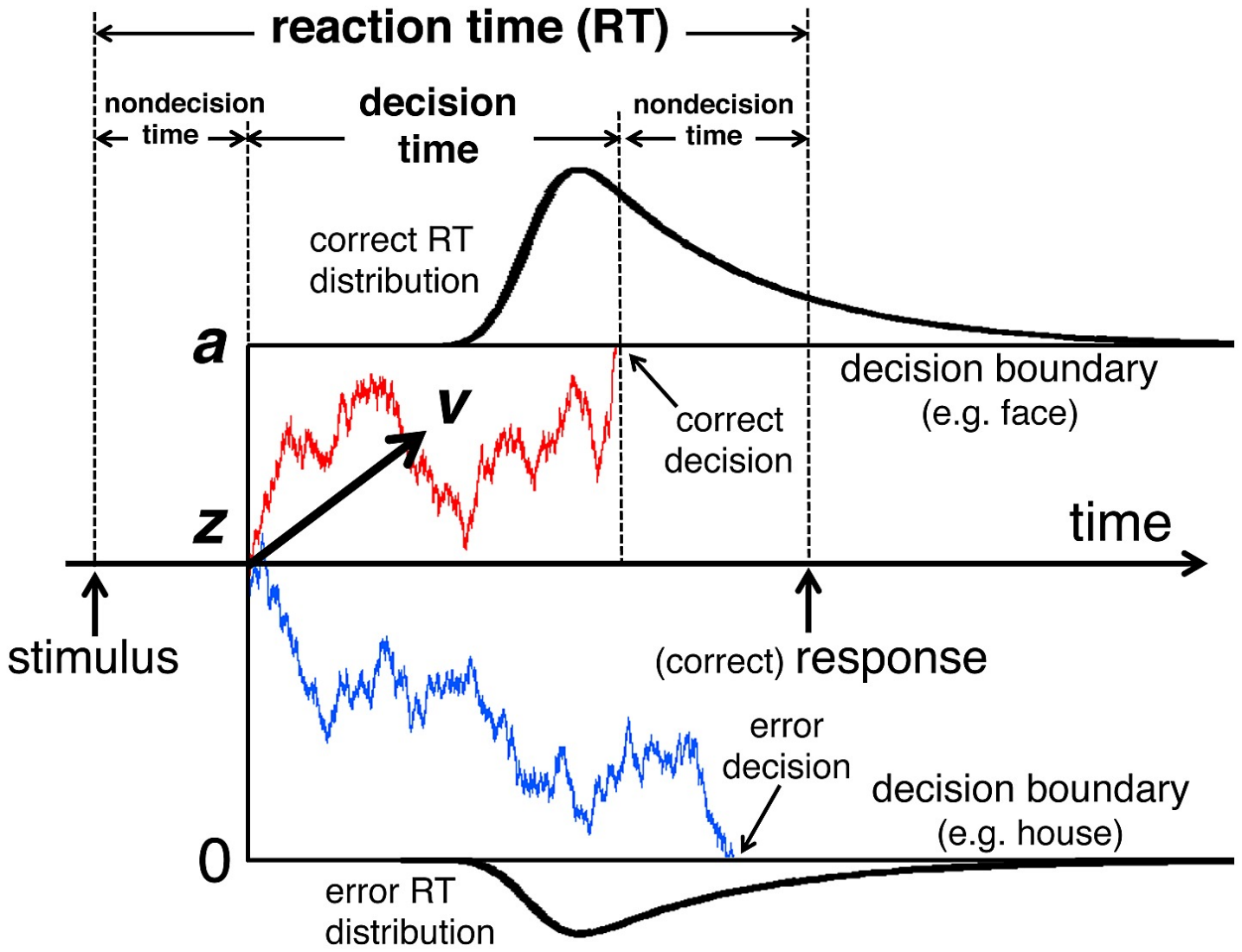
The ExGaussian Distribution: Model Comparison

```
##                      LOOIC    SE
## fit_rt1             341.39 30.79
## fit_rt2             269.59 23.21
## fit_rt3             266.02 24.00
## fit_rt1 - fit_rt2    71.80 21.09
## fit_rt1 - fit_rt3    75.37 22.42
## fit_rt2 - fit_rt3     3.57  5.11
```

The ExGaussian models clearly fit better

Additionally modeling σ and β does not improve model fit by much

The Wiener Diffusion Model: Introduction



The Wiener Diffusion Model: Example

```r
fit_wiener1 <- brm(rt | dec(resp) ~ grp, data = RT, wiener())
```

```
##  Family: wiener(identity)
## Formula: rt | dec(resp) ~ grp
##    Data: RT (Number of observations: 200)
## Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup samples = 4000
##     ICs: LOO = NA; WAIC = NA; R2 = NA
##
## Population-Level Effects:
##           Estimate Est.Error l-95% CI u-95% CI Eff.Sample Rhat
## Intercept    -0.06      0.11    -0.28     0.17       4000 1.00
## grpB          1.09      0.16     0.78     1.42       4000 1.00
##
## Family Specific Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Eff.Sample Rhat
## bs       1.96      0.07     1.82     2.10       3373 1.00
## ndt      0.31      0.02     0.27     0.34       3092 1.00
## bias     0.51      0.03     0.46     0.56       3395 1.00
##
## Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
## is a crude measure of effective sample size, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

The Wiener Diffusion Model: Predictions

You can predict upper bound responses as positive RTs and lower bound responses as 'negative' RTs

```
pred <- posterior_predict(
  fit_wiener1, newdata = RT[1:5, ],
  nsamples = 5, negative_rt = TRUE
)
round(pred, 2)
```

```
##       [,1]  [,2]  [,3]  [,4]  [,5]
## [1,] -0.82 -1.88 -1.16 -1.26  0.96
## [2,]  1.85  1.28  1.37  0.83  0.55
## [3,]  0.69 -1.54  0.47  0.73 -0.78
## [4,]  1.45 -1.54 -0.86 -1.25 -1.48
## [5,]  1.94 -1.03 -0.66  3.69  2.31
```

Time for exercise

'response_times_models.R'

Further Reading

- Wagenmakers E.-J. (2009). Methodological and empirical developments for the Ratcliff diffusion model of response times and accuracy. *European Journal of Cognitive Psychology*, 21(5), 641–671.
- Vandekerckhove J. and Tuerlinckx F. (2007). Fitting the Ratcliff diffusion model to experimental data. *Psychonomic Bulletin & Review*. 14(6), 1011–1026.